TriPoseFusion v2: Self-Calibrated Multi-View Fusion of Monocular 3D Keypoints for In-Cabin Driver Pose Estimation

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***Abstract*—Driver monitoring systems need stable 3D body pose under occlusion, constrained camera placement, and rapidly changing illumination. Off-the-shelf monocular 3D pose estimators can be deployed per camera, but their outputs live in inconsistent coordinate frames and exhibit view-dependent errors, and obtaining metric 3D ground truth inside a vehicle cabin is impractical. This paper presents a complete, self-calibrated pipeline for fusing synchronized tri-view monocular 3D keypoints. First, we show that a multi-view triangulation reference built with nominal camera extrinsics suffers per-session scale errors of up to $6\times$, and we introduce a per-sequence self-calibration procedure—robust bundle adjustment of the camera extrinsics from the 2D keypoint observations themselves, with the metric scale gauge anchored to the monocular estimator's body-size prior—that reduces the reference reprojection error from 10.3 to 3.6 px and its cross-session shoulder-width spread from 0.46–2.10 m to 0.368 ± 0.015 m without any calibration target or manual 3D annotation. Second, we present TriPoseFusion v2, a lightweight fusion network (28K parameters, <0.6 ms per frame on one CPU thread) that combines masked set attention over any subset of views, corruption-aware gating trained with view-corruption augmentation, a residual refinement head supervised by the self-calibrated triangulation reference, and a per-joint uncertainty output. On 88 subject-environment sequences under four illumination conditions and subject-disjoint 5-fold evaluation, TriPoseFusion v2 reduces MPJPE by 11.8% and hand error by 11% over the strongest training-free fusion baseline, with consistent gains in all four illumination conditions; when one camera fails, its accuracy degrades by less than 7% while assigning the corrupted view a weight of 0.01–0.03, matching an oracle that knows which view is corrupted. The predicted uncertainty correlates with the actual error (Spearman $\rho = 0.74$). As a downstream check, head yaw measured from the fused keypoints agrees with an independent head-rotation signal on every one of the 88 drives (worst case $r = 0.61$), whereas a single camera fails on up to four drives. On the public Drive&Act benchmark, the training-free canonicalized fusion transfers zero-shot at 23.1 mm PA-MPJPE, and re-running the adaptation recipe on the new rig—supervised only by triangulation of the rig's own observations, with no manual labels—improves it to 20.6 mm (18.6 mm when the dataset's own reference supervises, an upper bound in reference quality). We further contribute two cautionary analyses of broad relevance: a proof that consensus-teacher self-supervision collapses multi-view gating to uniform averaging, and a demonstration of how reference-scale errors manufacture illusory learning gains.**

***Index Terms*—Driver monitoring, 3D human pose estimation, multi-view fusion, self-calibration, intelligent vehicles.**

I. Introduction

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DRIVER monitoring and in-cabin behavior understanding rely on stable 3D representations of the driver's body. Fine-grained pose supports attention estimation, hand-state analysis, gesture understanding, safety intervention, and higher-level behavior recognition [1], [2], [3]. Compared with pose estimation in open scenes, the in-cabin setting is heavily constrained: the driver is seated, cameras are mounted at fixed but vehicle-specific positions, and safety-critical motion is concentrated in a small upper-body workspace. The same constraints create persistent visual challenges—the steering wheel, dashboard, seat, and the driver's own arms repeatedly occlude hands and torso landmarks; viewpoints are limited by the vehicle layout; and illumination changes sharply between day, night, and low-light scenarios.

Monocular 3D pose estimators [4], [5], [6], [7] are attractive here because they run independently per camera and need no reconstruction infrastructure. Their predictions, however, are depth-ambiguous and view-dependent: each camera returns keypoints in its own root-relative frame, with its own bias pattern, missing joints, and temporal jitter. In driver monitoring these limitations are most pronounced exactly where monitoring needs accuracy most—hands, wrists, and head landmarks affected by steering-wheel occlusion and motion blur. A multi-camera rig observes complementary evidence [8], [9], [10]: a joint invisible from the front may be visible from a side camera, and depth ambiguity in one view is often resolved by another. Exploiting this evidence at the *keypoint level*—the natural interface when the per-view estimator is an off-the-shelf component—raises three coupled problems that this paper addresses end to end.

(1) The reference problem. Learning and evaluating multi-view fusion requires a 3D reference, yet motion capture is impractical in a cabin, and even checkerboard calibration is difficult to repeat every time a rig is serviced or re-mounted. The common workaround, triangulating 2D keypoints with nominal (assumed) extrinsics, is dangerous in a way that is easy to miss: we show that on our 22-subject dataset the nominal-extrinsics reference has a median body-scale error of $2.4\times$, varying by recording session from $1.2\times$ to $6\times$ (Sec. III), because the rig geometry differed across sessions. All "metric" errors computed against such a reference are dominated by reference-scale error, not pose error—and, as we demonstrate, a learned model will happily absorb the reference's error and appear to improve.

(2) The fusion problem. Given well-aligned views, how much can a learned fusion module improve over simple

averaging, and where? We answer this under a strictly controlled protocol (identical inputs, references, validity masks, and metric code for every method) and find a nuanced picture: learned *view weighting* alone offers no accuracy gain—and we prove why its self-supervised variant cannot (Sec. IV-E)—whereas a small *residual refinement* head supervised by the self-calibrated reference gives consistent gains, concentrated on the hardest joints (hands).

(3) The robustness and transfer problem. A deployed system must tolerate a failed or corrupted camera, and a method developed on one rig must be transferable to another vehicle. We show that corruption-aware training makes gating match an oracle that knows which view failed, and that residual refinement is inherently rig-specific—it degrades under zero-shot transfer but is restored, without any manual labels, by re-running the same self-calibrated adaptation recipe on the target rig in minutes of CPU time.

This paper is a substantially revised and corrected successor of an earlier conference submission describing TriPoseFusion. In re-auditing that system we found that its headline improvements were artifacts of evaluation defects (an inverted Procrustes rotation and a degenerate canonicalization fallback that collapsed frames to the origin) and of the reference-scale error above. We document these findings as analyses (Secs. IV-E and IV-F) because they concern failure modes that can affect any work that trains or evaluates against pseudo ground truth.

Our contributions are:

- **A self-calibrated triangulation reference for in-cabin rigs.** Per-sequence robust bundle adjustment of camera extrinsics from the 2D keypoint observations themselves, with the scale gauge anchored to the monocular estimator's body-size prior. No calibration target, no manual 3D annotation. Reference reprojection error drops from 10.3 to 3.6 px, front-view utilization rises from 8% to 45%, and the cross-session shoulder-width spread collapses from std 0.37 m to 0.015 m (Tables I and II).
- **TriPoseFusion v2, a lightweight, deployment-oriented fusion network.** Masked set attention over the available views (any subset of 1–3 cameras), corruption-aware gating trained with view-corruption augmentation, residual refinement supervised by the self-calibrated reference, and a per-joint Laplace uncertainty output. 27.9K parameters; $<$0.6 ms per frame on a single CPU thread.
- **A controlled benchmark and an honest accounting of where learning helps.** Under subject-disjoint 5-fold evaluation, TriPoseFusion v2 improves MPJPE by 11.8%, hand error by 11%, and PCK@0.10 by $+4.1$ points over the strongest training-free fusion, with gains in all four illumination conditions; view weighting alone contributes nothing; under a corrupted camera, accuracy is maintained within 7% (vs. 19% degradation for averaging), matching an oracle exclusion; and a downstream head-orientation check shows the fused keypoints yield a valid head-yaw signal on all 88 drives, where single cameras fail on 3–4.
- **Cross-rig validation on Drive&Act** [11]: training-free canonicalized fusion transfers zero-shot (23.1 mm PA-MPJPE, in the range of fully supervised methods trained on that benchmark), and the fully label-free adaptation recipe—supervised only by DLT triangulation of the rig's own 2D keypoints—reduces it to 20.6 mm (leave-one-subject-out), with wrist error reduced by 24%; the dataset's own reference as supervision gives 18.6 mm, showing the gain scales with reference quality.
- **Two cautionary analyses**: a proof that consensus-teacher self-supervision makes uniform view weighting the loss-optimal solution (explaining "gate collapse" observations), and a quantified case study of how reference-scale errors manufacture illusory learning gains that do not transfer.

II. Related Work

A. Monocular and Video-Based 3D Pose Estimation

Monocular 3D pose estimation recovers joints by direct image-to-pose regression, 2D-to-3D lifting, or parametric body-model fitting [4], [5], [12], [6], [7]. Video-based models exploit temporal context to reduce per-frame ambiguity and jitter [13], [14]. These approaches are directly useful for driver monitoring because they can be deployed on each cabin camera independently. Single-view recovery nevertheless remains ill-posed under steering-wheel occlusion, truncation by the image border, viewpoint bias, and illumination extremes; the resulting 3D keypoints contain missing joints, inconsistent scale, incorrect depth ordering, and high-frequency jitter. We treat such outputs as noisy observations and study how to combine them across views, rather than proposing another monocular estimator.

B. Multi-View Pose Estimation and Keypoint-Level Fusion

Classical and learned multi-view methods triangulate 2D detections with calibrated cameras [15], [8], exchange image features or heatmaps across views [9], [16], or aggregate evidence volumetrically [10], [17], [18]. These methods assume image access and calibrated cameras at inference time, and volumetric aggregation carries a substantial computational cost that is difficult to justify on automotive edge hardware. Our interface is deliberately narrower: the fusion stage receives only per-view 3D keypoints from an existing estimator, requires no calibration at deployment (calibration is used only to construct the training reference), and runs in microseconds per frame. This keypoint-level setting exposes difficulties that image-level methods do not face—per-view root-relative frames that are mutually inconsistent, missing joints, and time-varying, joint-dependent view reliability—and, as we show, it changes which components actually matter: with well-normalized inputs, sophisticated cross-view weighting is worth little, while residual correction and robustness to view failure are worth a lot.

C. Self-Supervision, Pseudo Labels, and Their Failure Modes

When metric 3D labels are unavailable, prior work substitutes reprojection consistency, multi-view agreement, bone-length priors, temporal smoothness, or triangulated pseudo labels [19], [20], [8], [21], [22]. Our reference construction

belongs to this family but differs in two respects: extrinsics are refined *per recording session* directly from the human-keypoint correspondences (no calibration target), and the metric scale gauge—unobservable from uncalibrated multi-view geometry—is anchored to the monocular estimator's body prior and validated against anatomy. We further contribute two failure-mode analyses relevant to this entire line of work: consensus-teacher objectives provably drive adaptive view weighting to uniform averaging (Sec. IV-E), and scale errors in a pseudo reference are silently absorbed by refinement networks, inflating apparent gains that evaporate under cross-rig transfer (Sec. IV-F).

D. Driver Monitoring Datasets and In-Cabin Pose

In-cabin behavior datasets support distraction and activity recognition [1], [2], [3]; Drive&Act [11] additionally provides multi-view recordings with OpenPose-triangulated 3D references and serves as our cross-rig testbed. Our own dataset complements it with a three-camera RGB rig, 22 subjects, four illumination conditions, and a 52-keypoint protocol that includes both hands—the body parts that dominate the in-cabin error budget yet are frequently truncated or occluded.

III. Self-Calibrated Triangulation Reference

A. Rig, Estimator, and Triangulation Pipeline

Our rig has three cameras (front, left, right) observing the seated driver (Fig. 1). Each view runs the same off-the-shelf monocular estimator (SAM3D-Body), which outputs 70 2D and 3D keypoints per frame; a 52-keypoint subset (5 head, 5 shoulder/neck, and 2×21 hand keypoints) is used throughout. The reference is built by triangulating the synchronized per-view 2D keypoints [15]. For each joint and frame: every subset of ≥ 2 views with valid, in-bounds observations is triangulated by DLT; the subset with the lowest mean reprojection error is retained (ties broken toward more views); candidates with non-positive depth in any used camera are rejected; and points whose best mean reprojection error exceeds 40 px are marked invalid. Per-point quality statistics—multi-view and per-view reprojection errors, leave-one-view-out 3D displacement, and the DLT condition number—are stored alongside the reference, so that downstream users can filter or weight by reference confidence.

Intrinsics were estimated once per camera. Extrinsics are the crux, and the remainder of this section is about getting them—and the metric scale—right without a calibration target.

B. The Nominal-Extrinsics Problem

The extrinsics were originally *constructed* from a nominal layout: look-at poses derived from measured target distances and a 0.70 m left–right baseline. Reprojection statistics of the resulting reference look tolerable (mean 10.3 px; Table II), which is precisely why the defect below went unnoticed.

Auditing the reference against anatomy exposed it. The median triangulated shoulder width across the 88 sequences was 0.84 m—more than twice a plausible human value—and ranged from 0.46 m to 2.10 m depending on the *subject*, i.e., on the recording session (Table I, first row). The estimator's own metric body prior was, by contrast, extremely stable: median shoulder width 0.371 m with a spread below 0.06 m across all views, subjects, and illumination conditions. The scale error was independent of illumination but strongly session-dependent, implying that the rig was re-positioned between sessions while a single nominal geometry was assumed. Because the induced error is mostly scale-like, it poisons every metric error computed against the reference while remaining nearly invisible to reprojection-based quality checks: with a wrong baseline, reprojections stay consistent while the reconstruction is uniformly mis-scaled.

C. Per-Sequence Robust Extrinsic Refinement

We therefore refine extrinsics *per sequence*, using the 2D keypoint observations themselves as correspondences:

1) Fix the left camera as the gauge; parameterize the front and right cameras by 6-DoF perturbations of the nominal poses.
2) Sample $\sim$40 frames evenly from the sequence; collect all joints observed in ≥ 2 views.
3) Minimize the Huber-robustified reprojection error of DLT-triangulated points over the 12 camera parameters (variable-projection form: structure is re-triangulated inside the residual), with a soft prior toward the nominal layout.
4) Reject observations with residuals above 30 px against the round-1 solution and re-optimize.

Across 88 sequences the refinement rotates the nominal camera orientations by a median of 21° (front) and 31° (right)—up to 53°—confirming that the nominal layout was substantially wrong and different in every session. On held-out frames, the mean reprojection error drops from 9.6 to 3.4 px and the front camera's participation in accepted triangulations rises from 8% to 45%; the whole procedure takes well under a minute per sequence on a CPU.

D. Anchoring the Metric Scale Gauge

Rotation refinement alone cannot fix the scale. With unknown extrinsics, the global scale of the reconstruction is a gauge freedom—scaling the scene and the camera centers about the fixed camera leaves every reprojection unchanged—and pinning it with the nominal 0.70 m baseline is exactly what failed: back-solving from the refined geometry, the true baselines varied from 0.25 to 1.07 m (median 0.80 m) across sessions. We instead anchor the gauge *anatomically*: each sequence's reconstruction is rescaled about the fixed camera so that its median triangulated shoulder width equals the monocular estimator's median metric shoulder width for that same sequence. This choice (i) costs nothing in reprojection error, by gauge invariance; (ii) makes the reference's metric scale consistent with the very streams being fused, so that fusion errors measure pose, not scale disagreement; and (iii) is validated by convergence: the resulting shoulder widths concentrate at 0.368 ± 0.015 m across all 88 sequences, in agreement with both the estimator prior (0.371 m) and human anatomy. The anchor inherits any global bias of the

TABLE I
REFERENCE QUALITY ACROSS THE THREE EXTRINSICS VARIANTS (88 SEQUENCES, ∼0.7M FRAMES PER CAMERA). "SHOULDER WIDTH" IS THE PER-SEQUENCE MEDIAN OF THE TRIANGULATED LEFT–RIGHT SHOULDER DISTANCE; A PHYSICALLY CONSISTENT REFERENCE MUST CONCENTRATE NEAR 0.37 M.

Extrinsics	Valid joints	Mean reproj. (px)	Shoulder width (m)
Nominal layout	35.8%	9.60	0.839 [0.46, 2.10], σ=0.369
Global refinement	37.8%	5.18	0.410 [0.28, 3.85], σ=0.422
Per-seq. + anchor	37.7%	**3.31**	**0.368 [0.325, 0.389],** σ=0.015

TABLE II
REFERENCE REPROJECTION ERROR BY ILLUMINATION CONDITION AND BY VIEW, BEFORE (NOMINAL EXTRINSICS) AND AFTER (PER-SEQUENCE SELF-CALIBRATION WITH ANATOMICAL SCALE ANCHOR), ON THE VALID 52-KEYPOINT SUBSET. PER-VIEW ROWS ALSO SHOW THE NUMBER OF VALID OBSERVATIONS CONTRIBUTED BY THAT VIEW; SELF-CALIBRATION MULTIPLIES THE FRONT CAMERA'S USABLE CONTRIBUTION BY 4.7×.

		Nominal extrinsics				Self-calibrated			
Group	Subset	n	Mean	Med.	P95	n	Mean	Med.	P95
Overall	All	17.1M	10.28	8.40	26.05	17.4M	**3.57**	**1.60**	**13.90**
Env.	Day high	4.4M	10.19	8.25	26.31	4.4M	3.91	1.80	14.90
	Day low	4.2M	10.18	8.29	26.53	4.3M	3.50	1.50	13.75
	Night high	4.6M	10.73	8.99	25.26	4.7M	3.37	1.55	12.90
	Night low	4.0M	9.96	7.98	26.11	4.0M	3.49	1.45	14.00
View	Front	1.4M	15.46	11.55	42.05	6.6M	3.16	1.05	13.95
	Left	16.9M	9.96	8.30	24.60	13.9M	4.18	1.90	16.05
	Right	16.0M	10.16	8.20	26.35	14.4M	3.17	1.55	11.85

estimator's body prior; scale-invariant metrics (PA-MPJPE) are unaffected, and a single tape measurement of one true baseline would convert the anchor to an external metric scale if required.

Tables I and II summarize the outcome. A single globally refined extrinsics set (Table I, middle row) already halves the reprojection error but cannot fix sessions whose geometry differs from the calibration subset—its shoulder-width spread is even larger. Only the per-sequence procedure with anatomical anchoring yields a reference that is simultaneously low-error (3.6 px point-level mean, 65% lower than nominal), scale-consistent (std 1.5 cm), balanced across illumination conditions, and balanced across views: under nominal extrinsics the front camera contributed only 1.4M usable observations because its wrong pose inflated its reprojection error past the acceptance threshold; after self-calibration it contributes 6.6M at the lowest median error of the three views. All experiments below use this reference; Sec. IV-F quantifies what happens when the nominal-extrinsics reference is used instead.

IV. TRIPOSEFUSION V2

A. Problem Setting

At inference, TriPoseFusion v2 receives synchronized monocular 3D keypoint streams $P_v \in \mathbb{R}^{T\times J\times 3}$ from the available views $v \in \mathcal{V}$, $1 \le |\mathcal{V}| \le 3$, with J=52, and outputs a fused sequence P^{out} plus a per-joint uncertainty σ. No camera calibration and no images are needed at inference; the calibrated reference of Sec. III is used only as the training signal.

B. Robust Body-Centered Canonicalization

Each view is first mapped into a body-centered frame. With neck position o_t and shoulder positions s_t^l, s_t^r:

$$\mathbf{x}_t = \frac{s_t^r - s_t^l}{\|s_t^r - s_t^l\|}, \quad \mathbf{d}_t = \tfrac{1}{2}(s_t^l + s_t^r) - o_t, \tag{1}$$

$$\mathbf{z}_t = \frac{\mathbf{x}_t \times \mathbf{d}_t}{\|\mathbf{x}_t \times \mathbf{d}_t\|}, \quad \mathbf{y}_t = \mathbf{z}_t \times \mathbf{x}_t, \quad \bar{P}_v^{t,j} = R_t^\top (P_v^{t,j} - o_t), \tag{2}$$

with $R_t = [\mathbf{x}_t, \mathbf{y}_t, \mathbf{z}_t]$. Two robustness rules matter in practice. If a frame's shoulder estimate is degenerate—coincident points or an anatomically implausible width ($< \epsilon$ or > 1.0 m)—the shoulder axis of the *temporally nearest valid frame* is reused; substituting a zero vector (as an earlier implementation did) collapses R_t to rank zero and maps the entire frame to the origin. If $\mathbf{x}_t$ and $\mathbf{d}_t$ become near-parallel, a fixed reference axis restores full rank. Canonicalization is what makes per-view streams commensurable: it reduces the cross-view disagreement of raw monocular outputs from the 2 m scale (root-relative camera frames) to the 0.15 m scale on our data.

C. Masked Set Attention over Views

For each joint j and frame t, the $|\mathcal{V}|$ canonicalized observations form a token set. Each token is

$$u_v^{t,j} = W_{\text{in}}\big[\bar{P}_v^{t,j};\ \delta_v^{t,j};\ m_v^{t,j}\big] + e_v^{\text{view}} + e_j^{\text{joint}}, \tag{3}$$

where $\delta_v^{t,j} = \|\bar{P}_v^{t,j} - \mu^{t,j}\|$ is the distance to the finite-view mean $\mu^{t,j}$ (a cross-view-consistency cue), $m_v^{t,j}$ a validity flag, and $e^{\text{view}}, e^{\text{joint}}$ learned embeddings. Two transformer encoder layers (d=32, 4 heads) operate *across the view set* with key masking of invalid views. Attention over a set is permutation-equivariant [23], so the same network accepts any subset of cameras; a view that drops out at runtime is excluded exactly like a view that was never installed.

A linear head maps each attended token to a gate logit $g_v^{t,j}$; a masked softmax yields per-joint, per-frame view weights and the gated fusion:

$$\alpha_{t,j,v} = \frac{\exp g_v^{t,j}}{\sum_{v'\in\mathcal{V}_{t,j}} \exp g_{v'}^{t,j}}, \qquad \hat{p}_{t,j} = \sum_v \alpha_{t,j,v}\, \bar{P}_v^{t,j}. \tag{4}$$

The attention-pooled feature $h_{t,j} = \sum_v \alpha_{t,j,v} u_v^{t,j}$ is then processed by one transformer layer across the J joints (skeleton context, e.g., wrist evidence informing finger corrections) and a depthwise temporal convolution (kernel 5), and two zero-initialized heads output a residual correction and a Laplace log-scale:

$$p_{t,j}^{\text{out}} = \hat{p}_{t,j} + \Delta_{t,j}, \qquad \sigma_{t,j} = \exp(s_{t,j}). \tag{5}$$

The residual head is the component that can move joints *away* from all observations—for example, correcting a shared depth bias of the monocular estimator—which pure convex weighting cannot do.

D. Training: Reference Supervision, Corruption Augmentation, Uncertainty

The only loss is a masked Laplace negative log-likelihood against the self-calibrated reference $g_{t,j}$ over its valid joints Ω:

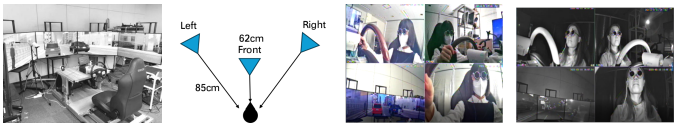
$$\mathcal{L} = \frac{1}{|\Omega|} \sum_{(t,j)\in\Omega} \Big(\|p^{\text{out}}_{t,j} - g_{t,j}\|_1 \, e^{-s_{t,j}} + 3\, s_{t,j} \Big), \qquad (6)$$

which trains the residual head and calibrates the uncertainty jointly [24]. During training, with probability 0.5 per sample, one randomly chosen view is corrupted: dropped (masked out), zeroed (a constant stream—an undetected sensor failure), or perturbed with σ=0.1 m Gaussian noise. This *corruption augmentation* is what teaches the gate to detect and down-weight unreliable views; without an external supervision signal and corrupted examples, the gate has no incentive to encode reliability (Secs. IV-E, V-D).

Implementation. Deployed capacity is d=32 (27.9K parameters). Training uses Adam (10^{-3}), 3,000 steps of batch 16 $\times$ 32-frame windows sampled from fp16-cached canonicalized sequences, a fixed seed, and evaluation of the final checkpoint only (no model selection on the evaluation fold). One fold trains in $\sim$8 minutes on four CPU threads; adaptation to a new vehicle is computationally trivial. Three deliberate simplicity choices follow from our analyses: no consensus self-supervision (Sec. IV-E); no contrastive view-alignment loss, which actively erases the inter-view differences the gate must read; and no extra capacity, which does not help and eventually hurts (Sec. V-H).

E. Why Consensus Self-Supervision Collapses Gating

The predecessor system (v1) trained its fused output against the per-frame coordinate-wise *median* of its own canonicalized inputs, with auxiliary self-referential terms (view consistency toward the inputs, bone lengths measured against its own pre-refinement output, temporal smoothness) and a contrastive view-alignment loss. Consider any fusion of the form $\hat{p} = \sum_v \alpha_v \bar{P}_v$ with α in the simplex, trained to minimize $\mathbb{E}\,\rho(\hat{p} - T(\bar{P}_{1:V}))$ where the teacher T is a symmetric central-tendency statistic (mean or median) of the same inputs. Under exchangeable per-view noise, the objective is symmetric in the views and its minimizer over α is the uniform weight $\alpha_v = 1/V$; no input pattern breaks the symmetry, because the teacher moves with the inputs. The prediction is architecture-independent and specific: gates converge to exactly $1/V$ and the output reduces to mean fusion. Empirically, every retrained v1 configuration—with or without cross-view attention, with different regularization weights, and with a leave-one-view-out teacher variant—converged to gates of $0.333/0.333/0.333$; under our unified protocol the v1 model scores 0.1857 m MPJPE versus 0.1858 m for plain mean fusion, and the collapse reproduced zero-shot on Drive&Act. Adaptive multi-view weighting requires information *external* to the views being weighted: in TriPoseFusion v2, the triangulation reference and the corruption augmentation.



(a) Simulator (b) Cameras (c) Daytime (d) Nighttime

Fig. 1. In-cabin acquisition setting: driving simulator, fixed tri-view camera rig, and representative daytime/nighttime frames illustrating illumination change, steering-wheel occlusion, and view-dependent visibility. Camera placement varied across recording sessions, which motivates the per-sequence self-calibration of Sec. III.

TABLE III
IN-CABIN DATASET COMPOSITION. EACH SUBJECT CONTRIBUTES ONE SEQUENCE PER ILLUMINATION SETTING.

Setting	Subjects	Sequences	Views / Keypoints
Day high	22	22	3 / 52
Day low	22	22	3 / 52
Night high	22	22	3 / 52
Night low	22	22	3 / 52
Total	22	88	3 / 52

F. How Reference-Scale Errors Manufacture Illusory Gains

Against the nominal-extrinsics reference (Table I, first row), a 1.9K-parameter fusion model with a temporal head appeared to beat mean fusion by 12.5% MPJPE, consistently across folds. The gain was an artifact. The reference is on average $2.4\times$ too large, so the network's temporal convolution learned an approximately global inflation of its output—the measured output scale was $2.1\times$ the input scale—which reduces error against a mis-scaled reference while corrupting the pose itself. The tell-tale signatures were: (i) the entire gain disappeared when the temporal head was removed; (ii) scale-sensitive and scale-invariant metrics moved in opposite directions; and (iii) zero-shot transfer to Drive&Act, whose reference has correct scale, *degraded* from 23.1 to 28–37 mm PA-MPJPE with a rigid-alignment error of 0.26 m. After scale anchoring, the same model's gain over mean fusion is 0.4%—zero. Pseudo-ground-truth supervision is widespread in pose estimation; this case study shows that a scale-consistent reference (or at minimum a scale-invariant metric) is a prerequisite for claiming learned improvements.

V. EXPERIMENTS

A. Datasets

In-cabin dataset (Fig. 1, Table III). 22 subjects, four illumination settings (day/night $\times$ high/low), one sequence per subject and setting: 88 sequences, $\sim$0.7M frames per camera. Each camera stream is processed by SAM3D-Body; the model input is the 52-keypoint subset covering the head, shoulders/neck, and both hands. The evaluation reference is the self-calibrated triangulation of Sec. III; metrics are computed over its valid joints ($\sim$17M joint observations).

Drive&Act [11]. Official split-0 *test* subjects (vp5, vp11, vp13; two runs each), three NIR views (inner mirror and both A-pillars). For zero-shot evaluation we sample 7,855 synchronized triplets at 1 fps ($\pm$21 ms tolerance) over $\sim$126 minutes; for adaptation we extract dense 30 fps clips (three 20 s segments per run; 10,640 frames). The reference is the

TABLE IV
MAIN COMPARISON ON THE IN-CABIN DATASET (5-FOLD, SUBJECT-DISJOINT; M). ALL ROWS SHARE IDENTICAL INPUTS, REFERENCE, MASKS, AND METRIC CODE. Δ IS THE PAIRED PER-FOLD MPJPE CHANGE VS. MEAN FUSION.

Method	Params	MPJPE ↓	PA-MPJPE ↓	PCK@0.10 ↑	Δ
Front only	–	0.172±.025	0.059±.011	0.459	+16%
Left only	–	0.161±.029	0.046±.011	0.473	+8%
Right only	–	0.152±.025	0.045±.010	0.477	+3%
Median fusion	–	0.152±.024	0.046±.010	0.485	+2%
Mean fusion	–	0.149±.026	0.044±.009	0.497	–
Best single (oracle)	–	0.145±.022	0.046±.012	0.488	−2.6%
Gating-only + aug.	1.9K	0.149±.026	0.044±.009	0.503	+0.4%
v1 backbone retrained	261K	0.136±.032	0.042±.009	0.535	−9.0%
Residual-only + aug.	8.0K	0.128±.031	0.040±.010	**0.548**	−14.3%
TriPoseFusion v2 (full)	27.9K	0.132±.031	0.042±.009	0.538	−11.8%

official `openpose_3d` (BODY-25, multi-view triangulation); knee/ankle confidences are zero in-cabin, so metrics use the body14/upper12 joint sets for the training-free evaluation and the joints shared with our 52-keypoint space for model evaluation (head, shoulders, wrists; the BODY-25 neck definition differs and is excluded from supervision).

B. Protocol, Metrics, and Baselines

Protocol. Subject-disjoint GroupKFold with 5 folds on the in-cabin dataset; leave-one-subject-out (3 folds) on Drive&Act. Every learned model trains for a fixed number of steps with a fixed seed and is evaluated at the final step. All methods—training-free and learned—consume identical cached canonicalized inputs, the identical reference, identical validity masks, and identical metric code; this "same-path" discipline matters, since we found that mixing two canonicalization implementations between method and baseline rows can alone fabricate double-digit relative differences on hand joints.

Metrics. MPJPE; PA-MPJPE using the correct least-squares similarity solution [25] (an inverted rotation in a previous implementation systematically inflated PA errors—for well-aligned shapes a transposed Procrustes rotation can even exceed the unaligned error, which is how the defect was originally noticed); PCK@0.10 m; joint-group breakdowns (head / shoulders–neck / body / hands); all as mean ± std over folds. PA-MPJPE and MPJPE are computed over exactly the same joint mask.

Baselines. Per-view single baselines; the per-sequence *oracle* best single view (a diagnostic upper bound for view selection, not a deployable method); training-free canonicalized mean and median fusion; a gating-only learned model (1.9K parameters; view weighting without a residual head); and the v1 backbone (per-view encoder, cross-view attention, gate, dilated TCN) retrained with our supervision at two capacities. External image-based multi-view estimators are not comparable under this keypoint-level protocol (different inputs and supervision); the Drive&Act experiments provide the external anchor instead.

C. Main Results on the In-Cabin Dataset

Table IV gives the central accuracy picture; three observations. *First*, canonicalized mean fusion is a strong baseline: it beats every individual camera and sits within 3% of the per-sequence oracle camera, so on a healthy rig the clean-accuracy headroom for view *selection* is small. *Second*, learned view weighting adds nothing (gating-only: +0.4%; its learned weights are near-uniform, 0.32/0.33/0.35)—consistent with Sec. IV-E and with the small selection headroom, and now confirmed even under reference supervision. *Third*, residual refinement is the one real accuracy lever: the 8K residual-only model improves MPJPE by 14.3% and the full model by 11.8%, both surpassing the oracle single view; the retrained v1 backbone confirms the same supervision works on the older architecture (−9.0%) but with 10× more parameters and less robustness. Per joint group (Table V, top), the improvement concentrates where monitoring needs it most—hands (0.234 → 0.208, −11%) and head (−10%)—with shoulders/neck marginally degraded; these near-origin joints define the canonical frame and already sit at 2 cm. Per illumination (Table V, bottom), the gain is uniform across all four conditions (−9% to −16%), largest at night-high; fusion accuracy itself is stable from day to night, supporting the multi-view premise under illumination change.

TABLE V
MPJPE BY JOINT GROUP AND BY ILLUMINATION CONDITION (5-FOLD MEAN, M).

Method	Head	Shoulders/neck	Body	Hands
By joint group				
Mean fusion	0.081	**0.022**	0.052	0.234
Best single (oracle)	0.080	0.023	0.051	0.234
TriPoseFusion v2 (full)	**0.073**	0.025	**0.049**	**0.208**

	Day high	Day low	Night high	Night low
By illumination (MPJPE / PA-MPJPE)				
Mean fusion	.151/.042	.165/.056	.154/.041	.129/.039
TriPoseFusion v2 (full)	**.134/.039**	**.151/.053**	**.129/.038**	**.116/.039**
ΔMPJPE	−11%	−9%	−16%	−10%

D. Robustness to Camera Corruption

Table VI evaluates the failure mode that motivates multi-camera rigs: one view becomes unusable at runtime. Three corruption types are applied to each view in turn on the evaluation folds: *drop* (the view is absent), *zero* (the stream returns a constant pose—an undetected failure), and *noise* (σ=0.1 m). Mean fusion degrades by 19% under zeroing, because it ingests the corrupted stream at fixed weight 1/3. TriPoseFusion v2 assigns the zeroed view a weight of 0.01–0.03 and keeps its error within 7% of its clean value—*better* than the oracle that excludes the corrupted view, because the residual corrections remain active. The no-augmentation ablation shows corruption augmentation is the necessary ingredient: without it the model actually *up-weights* corrupted views (weight 0.31) and falls below the averaging baseline. Gaussian noise is harder to flag than zeroing (weights 0.09–0.38 depending on the view), which we identify as a limitation.

E. Cross-Rig Evaluation on Drive&Act

Drive&Act provides an independent rig, an independent reference (multi-view OpenPose triangulation, i.e., a source unrelated to our estimator), and unseen subjects.

TABLE VI
SINGLE-VIEW CORRUPTION AT TEST TIME (5-FOLD MEAN OVER THE THREE CORRUPTED VIEWS; MPJPE, M). "w" IS THE MEAN GATE WEIGHT ASSIGNED TO THE CORRUPTED VIEW (UNIFORM = 0.33). ORACLE REMOVES THE CORRUPTED VIEW BY FIAT.

Method	Clean	Drop	Zero (w)	Noise (w)
Mean fusion	0.149	0.152	0.176 (0.33)	0.167 (0.33)
Oracle excl. view	–	0.152	0.152 (0)	0.152 (0)
TriPoseFusion v2 w/o aug.	0.132	–	0.178 (0.31)	–
TriPoseFusion v2 (full)	**0.132**	**0.136**	**0.137 (0.02)**	**0.137 (0.22)**

Zero-shot, training-free. Running the same monocular estimator per view and our canonicalize-and-average fusion on the 1 fps triplets yields, pooled over 6,673 frames: body14 MPJPE 55.8 mm / PA-MPJPE 31.3 mm, upper-body-12 53.6/23.8 mm; fusion beats the best single view by 8% (body14 PA) to 13% (upper12 PA), and PCK@50 mm is 42% (94% @100 mm). For calibration of expectations: fully supervised methods trained on Drive&Act report 22.6–30.4 mm MPJPE; our numbers come from a system that has never seen this dataset, requires no calibration, and fuses in microseconds. On the shared-joint protocol used for our models (dense clips), the training-free fusion attains 23.1 mm PA-MPJPE (single views: 26.9/31.9/24.0; median: 23.5), and the gating-only model transfers unchanged (23.1 mm).

Calibrated triangulation baseline. Drive&Act also lets us report what calibrated geometry achieves on the same inputs—and, uniquely, *absolute* errors in the dataset's metric frame. Direct DLT triangulation of the per-view SAM3D 2D keypoints (undistorted, with the same 40 px acceptance filter as Sec. III) attains 32.5 mm body14 MPJPE (upper12: 22.2 mm; PA 29.0/18.0 mm) with no alignment of any kind—at the edge of the fully supervised range; the median residual between projected reference and detected 2D is 15.5 px, confirming calibration and synchronization. Removing any one view raises the error by 10–40% (worst pair, front+right: 45.5 mm), quantifying the third camera's contribution. Triangulation, however, requires calibrated extrinsics at inference time, which our calibration-free fusion does not; in the canonical frame the two are close in PA terms (29.0 vs. 31.3 mm body14).

Zero-shot, learned residuals do not transfer. Models with residual heads degrade under zero-shot transfer: 28.6 mm (full model), 36 mm (residual-only), 34.9 mm (retrained v1 backbone). The corrections they learned encode rig-and-estimator-specific systematic error, and the learned view priors even invert across rigs (front is the least-trusted view on our rig, $\alpha_f \approx 0.14$, but the most trusted after adaptation on Drive&Act, $\alpha_f \approx 0.40$–0.51): view reliability is a property of the installation, not of the method.

Label-free in-domain adaptation. The correct deployment recipe is therefore to re-run the adaptation on the target rig. In the fully label-free version, supervision is the DLT triangulation of the rig's own 2D keypoints (undistorted, 40 px acceptance filter—the same recipe as Sec. III); the official reference is used only for evaluation. Under leave-one-subject-out (Table VII), this reaches 20.6 mm PA-MPJPE for the residual-only model (-15% vs. training-free fusion) and 20.9 mm for the full model (-14%), with wrist error reduced by 19–24%. Supervising with the dataset's own reference instead—an upper bound on reference quality—yields 18.6 mm (-23%) and 19.9 mm (-18%); the gap is consistent with the self-built reference's own agreement with the official one (14–22 mm PA per run), i.e., the adaptation gain grows with reference quality. The recipe is also data-cheap: one minute of footage already yields -11% PA (two minutes: -20%; all four training sequences: -23%; dataset-reference supervision). Under a zeroed camera the adapted full model degrades to 24 mm while averaging degrades to 84 mm.

TABLE VII
DRIVE&ACT, LEAVE-ONE-SUBJECT-OUT ADAPTATION (3 FOLDS; SHARED UPPER-BODY JOINTS; M). "RIG REF." ROWS ARE SUPERVISED *only* BY DLT TRIANGULATION OF THE RIG'S OWN 2D KEYPOINTS (FULLY LABEL-FREE); "DATASET REF." ROWS USE THE OFFICIAL REFERENCE FOR SUPERVISION (AN UPPER BOUND IN REFERENCE QUALITY). ALL ROWS ARE EVALUATED AGAINST THE OFFICIAL REFERENCE.

Method	MPJPE ↓	PA-MPJPE ↓	Wrists ↓
Best single view (right)	0.0343±.003	0.0261±.003	0.071
Median fusion	0.0266±.002	0.0251±.002	0.041
Mean fusion	0.0247±.002	0.0243±.002	0.035
Gating-only + aug.	0.0235±.002	0.0228±.002	0.033
Residual-only (rig ref., label-free)	0.0225±.003	0.0206±.003	0.027
TriPoseFusion v2 (full, rig ref., label-free)	0.0228±.004	0.0209±.004	0.028
Residual-only (dataset ref.)	0.0203±.002	0.0186±.002	**0.026**
TriPoseFusion v2 (full, dataset ref.)	0.0217±.002	0.0199±.002	0.030

F. Uncertainty Calibration

The Laplace scale output orders errors well: the Spearman correlation between predicted scale and realized per-joint error is 0.74 on the in-cabin folds (stable at 0.71–0.74 across all architecture ablations) and 0.43–0.51 on adapted Drive&Act (label-free / dataset-reference supervision). Thresholding by confidence is directly useful downstream: the most-confident decile of joints carries 1.6 cm mean error versus 27 cm for the least-confident decile (14.8 cm overall); on adapted Drive&Act, 1.1 cm versus 3.4 cm. A monitoring stack can thus suppress unreliable joints per frame—a capability with no training-free counterpart.

G. Downstream Utility: Head-Orientation Measurement

To verify that pose-level gains translate into a DMS-relevant signal, we evaluate head orientation—the quantity behind mirror checks, glance behavior, and attention estimation—on all 88 in-cabin drives, with identical measurement code for every keypoint source. Head yaw is computed from the five head keypoints (nose, eyes, ears) in the canonical body frame; a head turn is a $|\text{yaw}| > 15^\circ$ excursion lasting ≥ 0.2 s. As an independent check signal we use the per-view estimator's head rotation matrix (a different output head than the keypoints), brought into the cabin frame through the self-calibrated extrinsics and averaged over the three views; it does not depend on nose/ear visibility. Human glance annotations (three raters) are deliberately *not* used as a head-turn reference: they mark eye-movement events, only 9% of which reach a 15° head rotation, and the keypoint yaw during labelled left/right glances deviates

TABLE VIII
HEAD-YAW MEASUREMENT VALIDITY ON THE 88 IN-CABIN DRIVES: PER-DRIVE PEARSON CORRELATION BETWEEN KEYPOINT YAW AND THE INDEPENDENT HEAD-ROTATION SIGNAL (MEAN / MINIMUM OVER DRIVES), NUMBER OF DRIVES WHERE THE MEASUREMENT FAILS ($r < 0.5$), AND THE RESULTING HEAD-TURN RATE. SAME CODE AND THRESHOLDS FOR EVERY SOURCE.

Keypoint source	r mean	r min	Failed drives	Turns/min
Front only	0.76	0.16	4	2.44
Left only	0.83	0.52	0	2.38
Right only	0.80	0.21	3	2.29
Median fusion	0.83	0.44	1	2.20
Mean fusion	**0.86**	0.50	1	2.12
Residual-only + aug. (8K)	**0.86**	0.56	**0**	2.28
TriPoseFusion v2 (full)	**0.86**	**0.61**	**0**	2.53

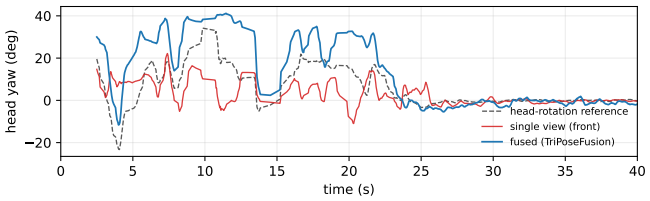


Fig. 2. Head yaw over a 40 s window of one night-time, low-illumination drive (subject 17): the front camera alone (red) halves the amplitude of the large turns and misses the turn around t=20 s, while the fused keypoints (blue) track the independent head-rotation reference (dashed), with some over-estimation of the largest excursions; window correlation with the reference 0.23 vs. 0.82.

from the resting yaw by $< 0.5°$ on average with chance-level direction agreement (47–50% for every source)—i.e., they carry no head-orientation information.

Table VIII shows that single cameras are not a reliable source of head orientation: the front and right cameras each fail outright on 3–4 drives (r as low as 0.16), the characteristic signature of steering-wheel and self-occlusion of the ears from one viewpoint. Training-free fusion already removes most failures, and the learned models remove all of them while raising the worst-case correlation to 0.61—the real-occlusion counterpart of the synthetic corruption results in Sec. V-D. The measurement itself is stable across illumination: the head-turn rate, peak angle, and yaw spread do not differ significantly between the four lighting conditions (Friedman test, $p > 0.4$), consistent with the illumination breakdown of Table V. Fig. 2 illustrates the typical failure: the front view systematically under-estimates turn amplitude and drops entire turns. Two caveats apply. The validation targets the head signal; utility for hand-related functions is supported only indirectly, by the per-joint hand improvements. And the check signal, although a separate output head and multi-view consistent (8–22° across views), comes from the same estimator family and is therefore only partially independent.

H. Ablations, Capacity, and Efficiency

Table IX isolates the contributions. Removing corruption augmentation leaves clean accuracy unchanged but destroys robustness (Sec. V-D); removing the skeleton-attention layer or the temporal convolution changes accuracy by $\leq 0.3\%$—consistent with the residual head being the active ingredient—

TABLE IX
ABLATIONS (5-FOLD; IN-CABIN DATASET; M).

Variant	Params	MPJPE	PA-MPJPE	Zero-corr. MPJPE
TriPoseFusion v2 (full)	27.9K	0.1320	0.0423	**0.137**
– corruption aug.	27.9K	0.1324	0.0430	0.178
– skeleton attention	19.4K	0.1323	**0.0408**	–
– temporal conv.	27.7K	0.1323	0.0424	–
Residual-only + aug.	8.0K	**0.1284**	0.0402	**0.132**
Residual (large)	92K	0.1311	0.0435	–
v1 backbone (h64) + aug.	261K	0.1359	0.0416	0.138
v1 backbone (h64) + velocity feats.	262K	0.1347	0.0412	–
v1 backbone (h128, original capacity)	870K	0.1423	0.0437	–

so deployments free to drop them can use a 19.4K variant. Capacity beyond 8–28K parameters does not help: a 92K residual variant is flat, and the v1 backbone at its original capacity (870K) is the *worst* learned model (-4.2% on average, with a per-fold range of -21% to $+26\%$), overfitting 68 training sequences. Multi-scale velocity features, a v1 component, contribute at most 0.6% when re-added. On efficiency: the full model runs in $<$0.6 ms per frame on one CPU thread (residual-only: 0.08 ms; both fit in $<$1 MB), one adaptation fold trains in $\sim$8 CPU-minutes, and no GPU is required anywhere in the fusion pipeline.

VI. DISCUSSION AND LIMITATIONS

Our results reframe what a learned in-cabin fusion module should claim. On a healthy three-camera rig, training-free canonicalized averaging is nearly optimal for clean accuracy; the honest value of learning lies elsewhere—robustness to camera failure (at oracle level, at zero cost to clean accuracy), rig-specific residual correction (worth 12–15% when retrained on the target rig's own automatic reference, and up to 23% as reference quality improves), and calibrated per-joint confidence. The same results carry a warning: without the reference repair of Sec. III, we would have reported a fictitious learning gain built on reference-scale error, and without the collapse analysis of Sec. IV-E we would have kept attributing meaning to a gate that mathematically cannot deviate from uniform.

Limitations remain. The in-cabin reference is triangulated from the same 2D detections that feed the fused streams; Drive&Act's independent OpenPose-based reference partially fills this gap, but a physically independent sensor would strengthen absolute claims. Hands still dominate the error budget (0.21 m vs. 0.05 m for the body) and are at times invisible to all three cameras—a sensing limit that no fusion can overcome. Gaussian corruption is only partially detected by the gate. The scale anchor inherits any global bias of the monocular estimator's body prior (PA-MPJPE is unaffected; one physical baseline measurement would externalize the scale). Finally, our data comes from a driving simulator; validating the self-calibration procedure on moving vehicles with vibration is future work.

VII. CONCLUSION

We presented a self-calibrated pipeline for multi-view driver pose estimation at the keypoint level: a per-sequence extrinsic refinement with anatomically anchored scale that turns

nominal-extrinsics triangulation into a physically consistent reference (3.6 px mean reprojection error; 0.368 ± 0.015 m shoulder width; $4.7\times$ more usable front-view observations), and TriPoseFusion v2, a 28K-parameter set-attention fusion network with corruption-aware gating, reference-supervised residual refinement, and calibrated uncertainty. The system improves accuracy by 11.8% over strong training-free fusion with gains in all four illumination conditions, tolerates a failed camera at oracle level, transfers across rigs via minutes of label-free adaptation supervised only by the rig's own triangulated observations (20.6 mm PA-MPJPE on Drive&Act), and runs in microseconds per frame on a CPU. The accompanying analyses—consensus self-supervision provably collapses view gating, and reference-scale errors manufacture illusory gains—are cautionary results we believe generalize well beyond driver monitoring.

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